PROYECTO INTERMODULAR

Informe

Simulación en Cisco Packet Tracer

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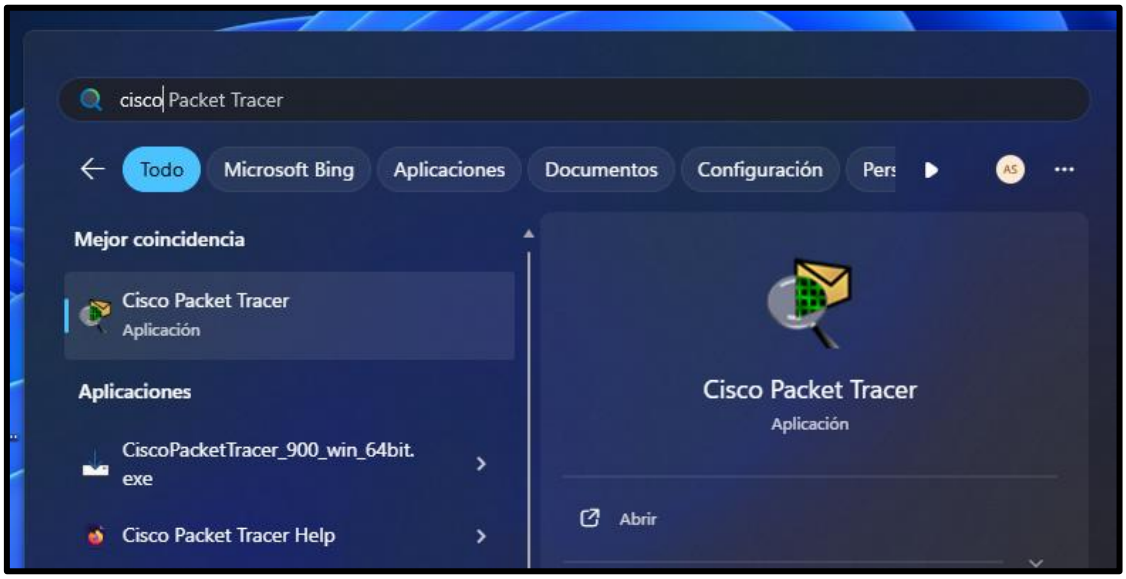
Configuración Inicial

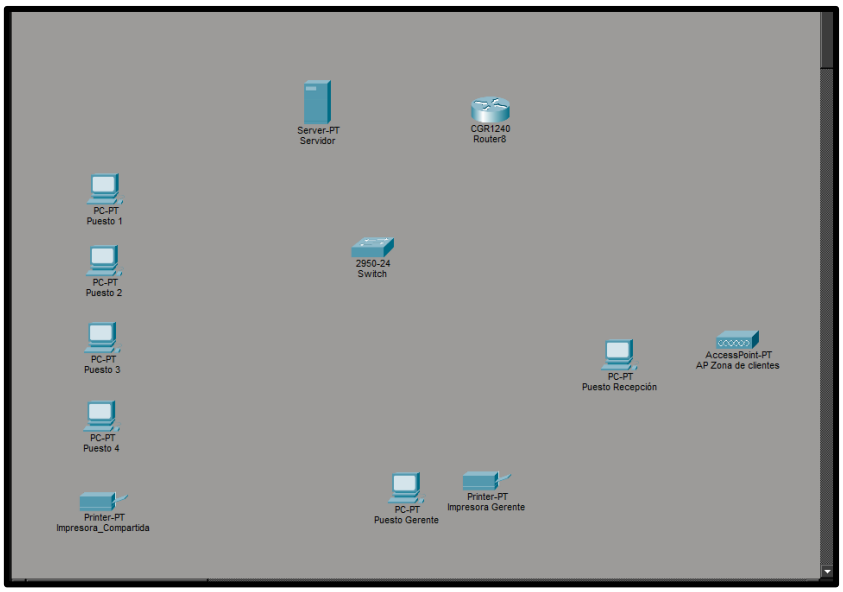
Abrimos Cisco Packet Tracer:



Agregamos los dispositivos que formaran parte de la red:

Hosts:

- Servidor
- Router
- Switch



VLAN 10:

- Equipo del Puesto 1
- Equipo del Puesto 2
- Equipo del Puesto 3
- Equipo del Puesto 4
- Impresora Compartida

VLAN 20:

- Equipo del Gerente
- Impresora del Gerente

VLAN 30:

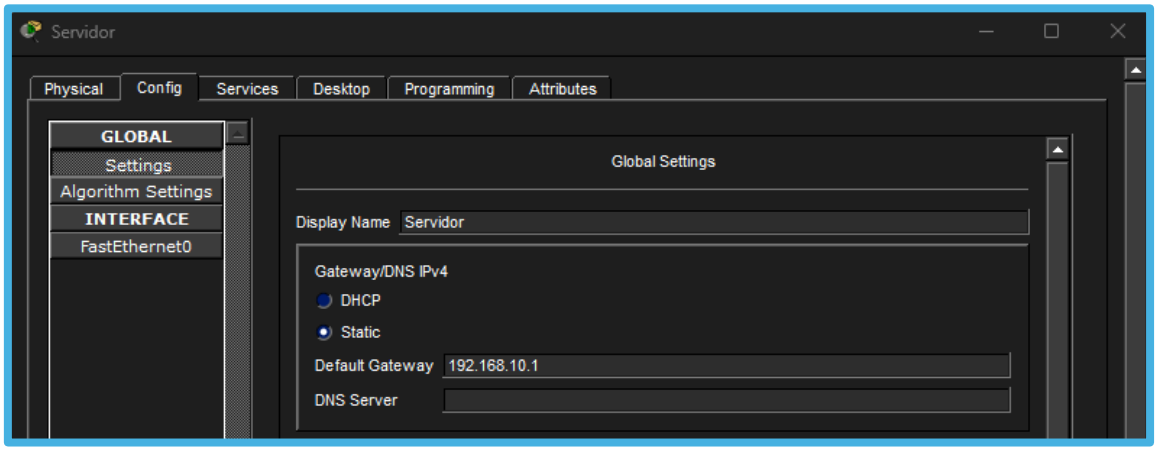
- Equipo del puesto de Recepción

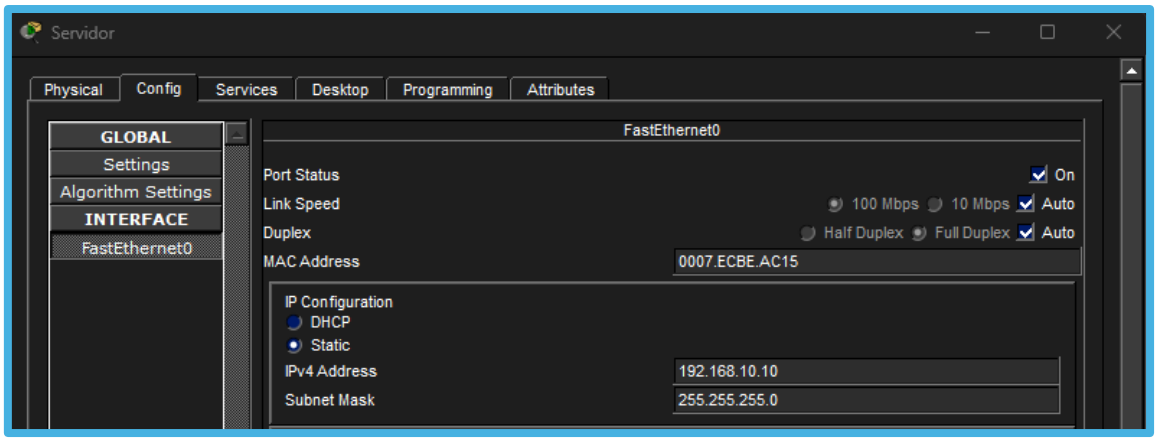
VLAN 40:

- Punto de Acceso para clientes

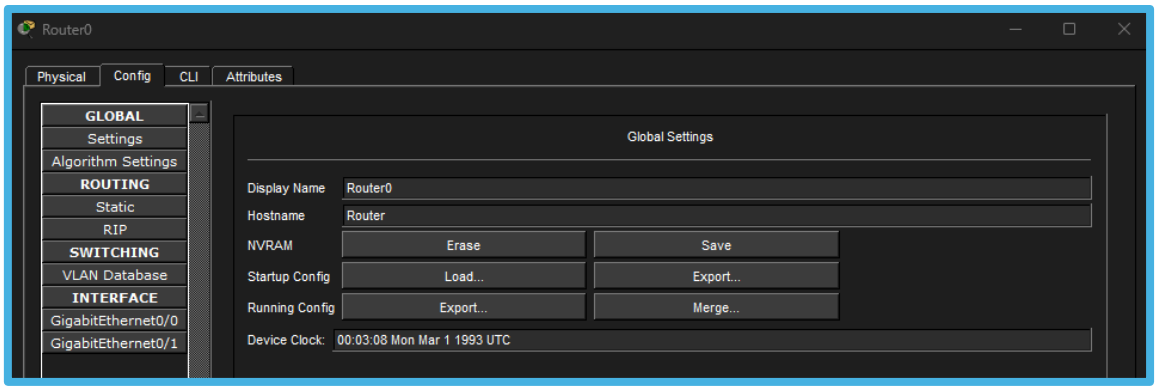
Hosts

Servidor:

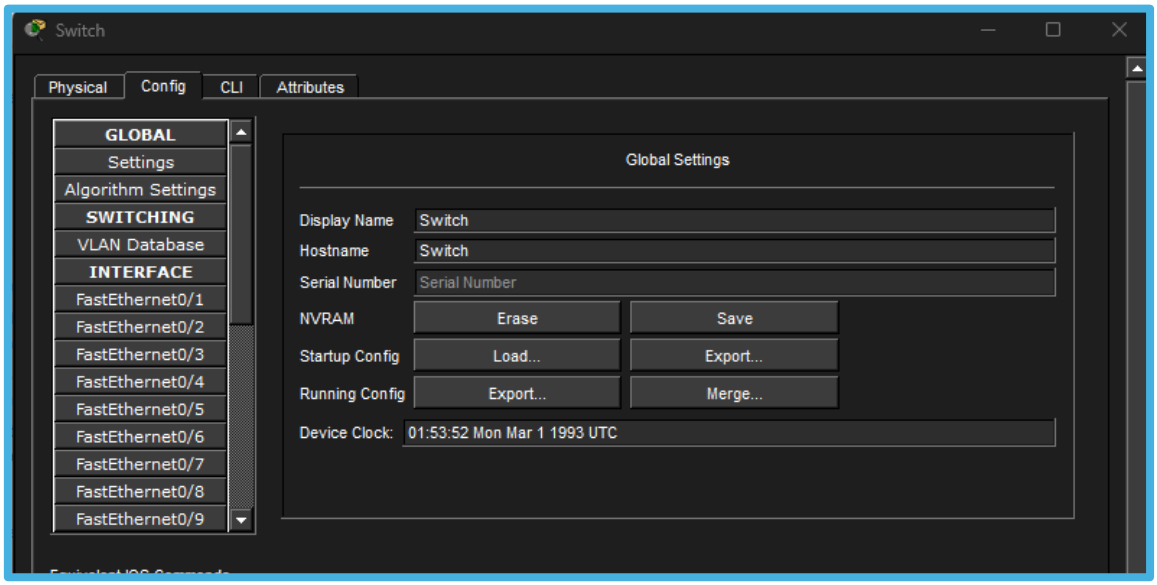


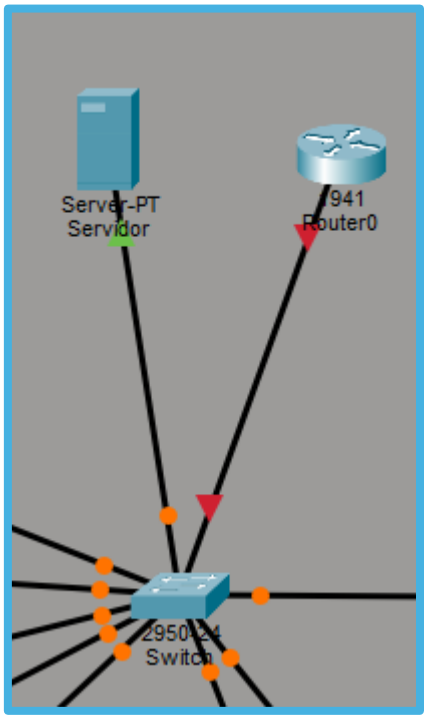


Router:



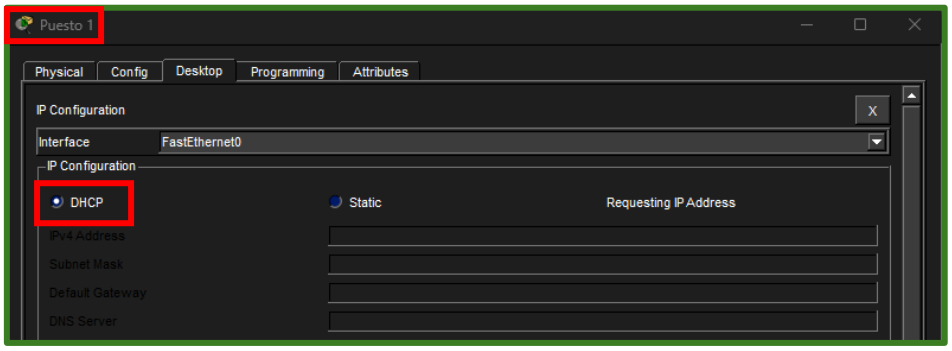
Switch:


Switch
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/1
FastEthernet0/2
FastEthernet0/3
FastEthernet0/4
FastEthernet0/5
FastEthernet0/6
FastEthernet0/7
FastEthernet0/8
FastEthernet0/9
Global Settings
Display Name
Switch
Hostname
Switch
Serial Number
Serial Number
NVRAM
Erase
Save
Startup Config
Load...
Export...
Running Config
Export...
Merge...
Device Clock:
01:53:52 Mon Mar 1 1993 UTC


Server-PT
Servidor
941
Router0
2950-24
Switch

VLAN 10

Equipo del Puesto 1:



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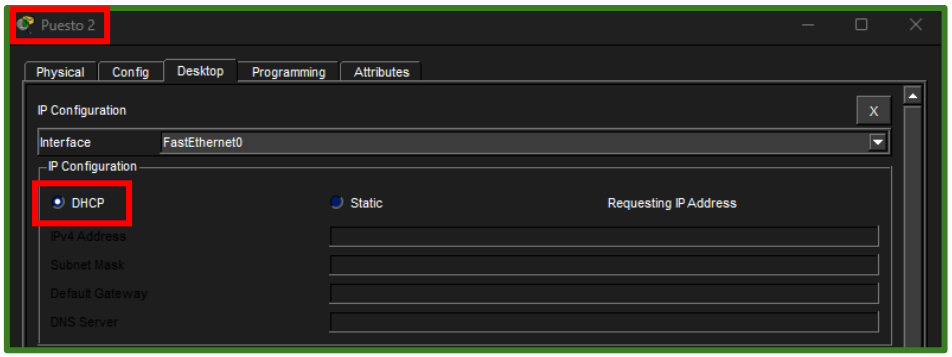
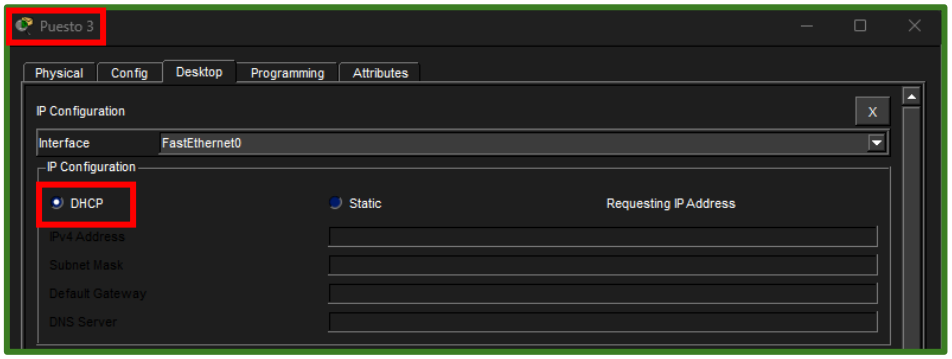
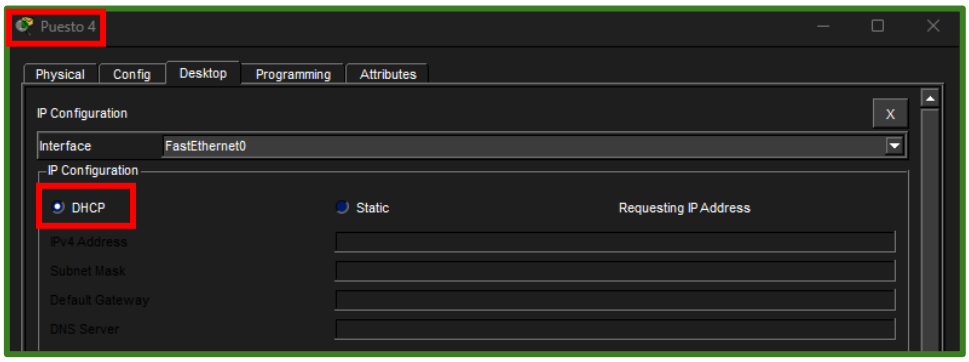


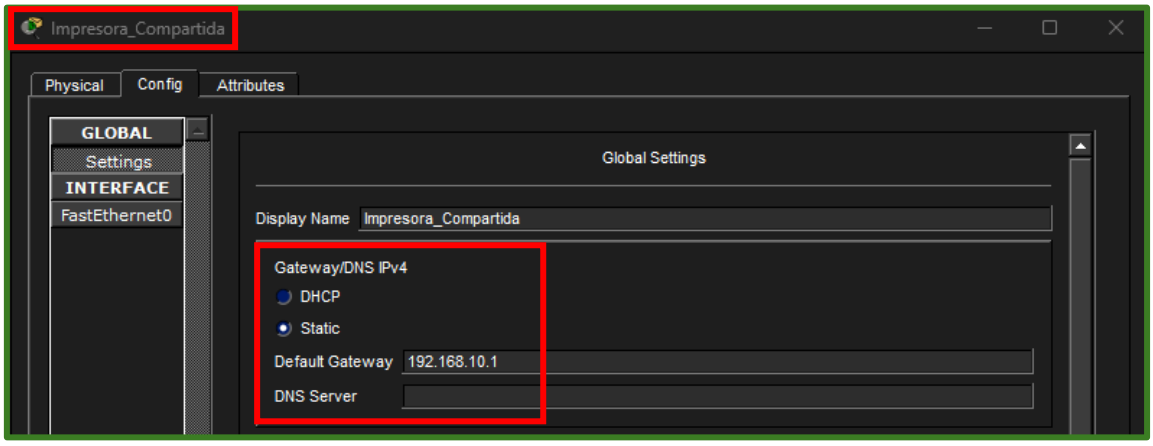
Equipo del Puesto 3:

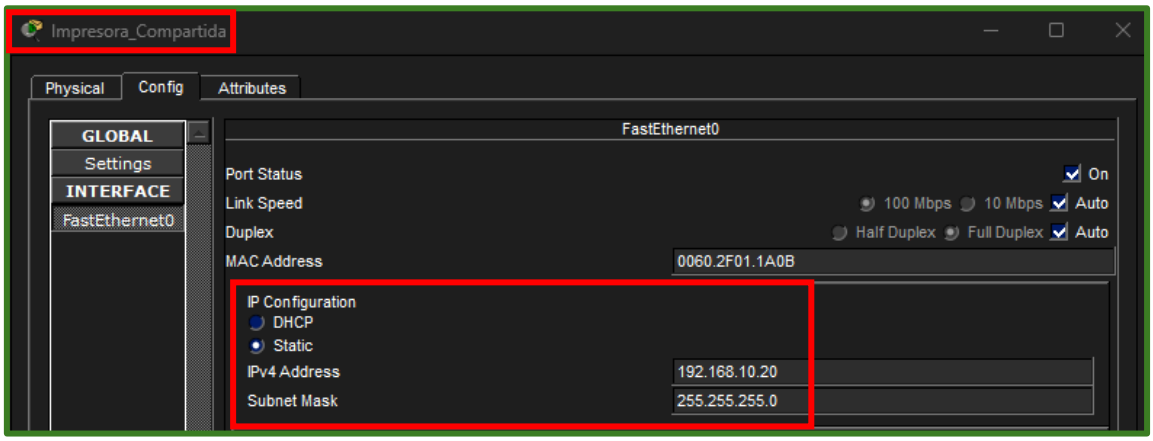


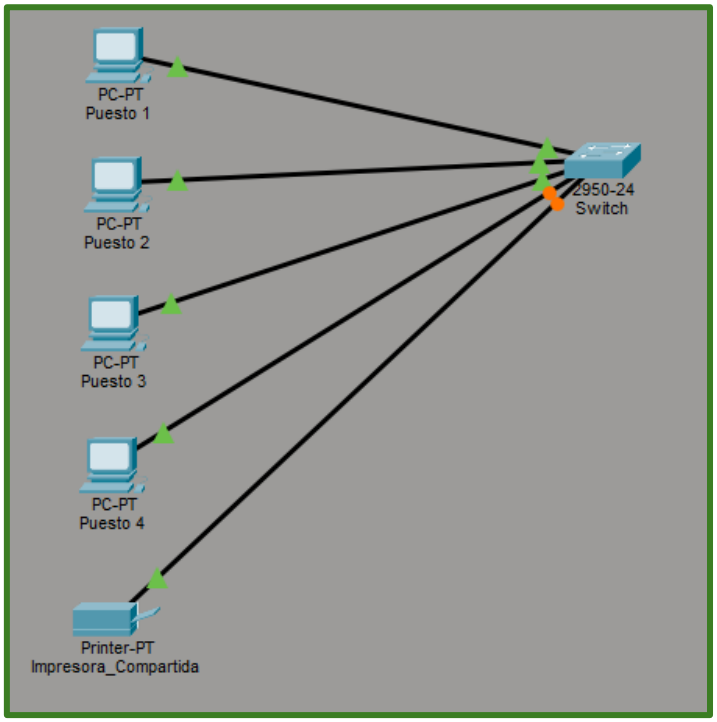
Equipo del Puesto 4:



Impresora Compartida:

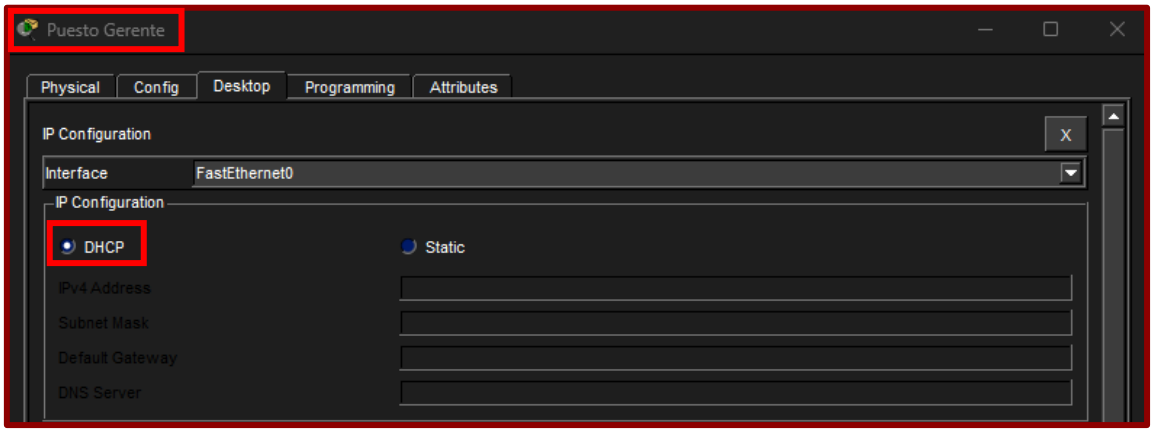




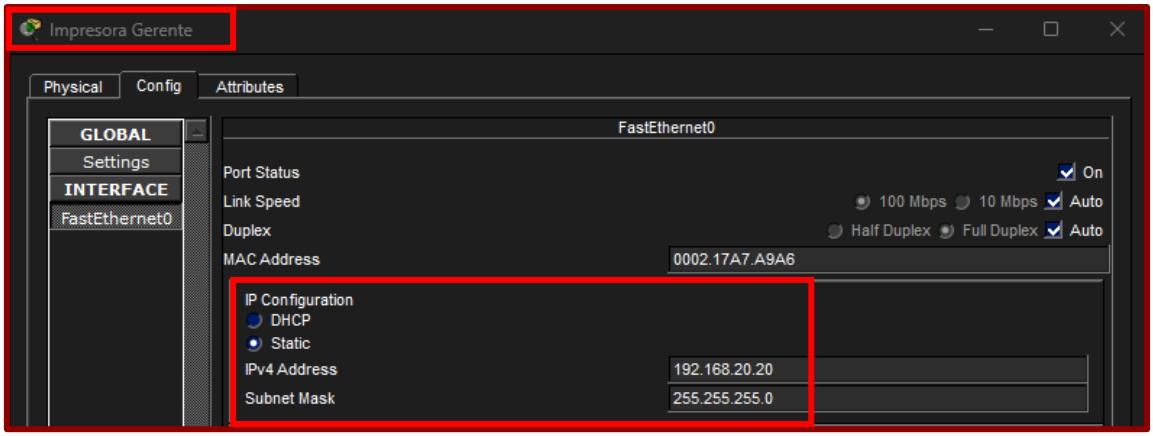


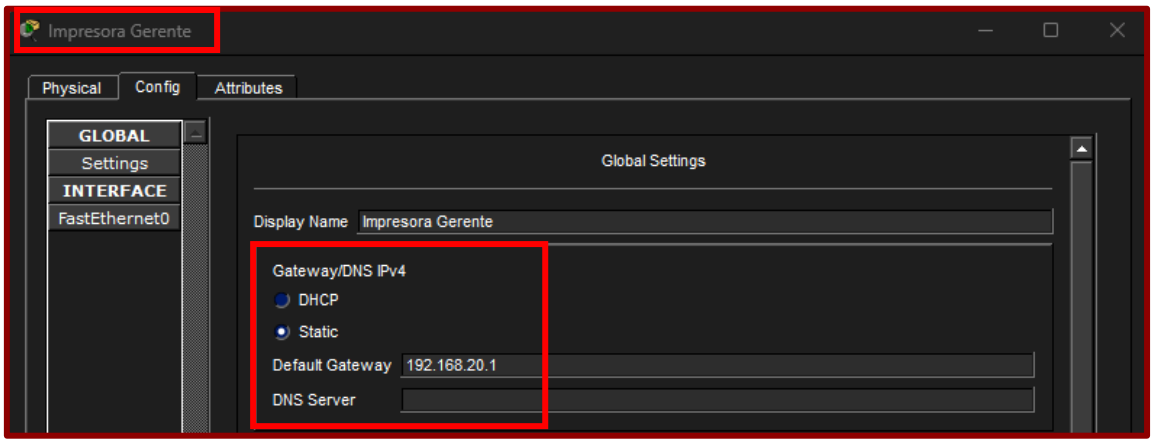
VLAN 20

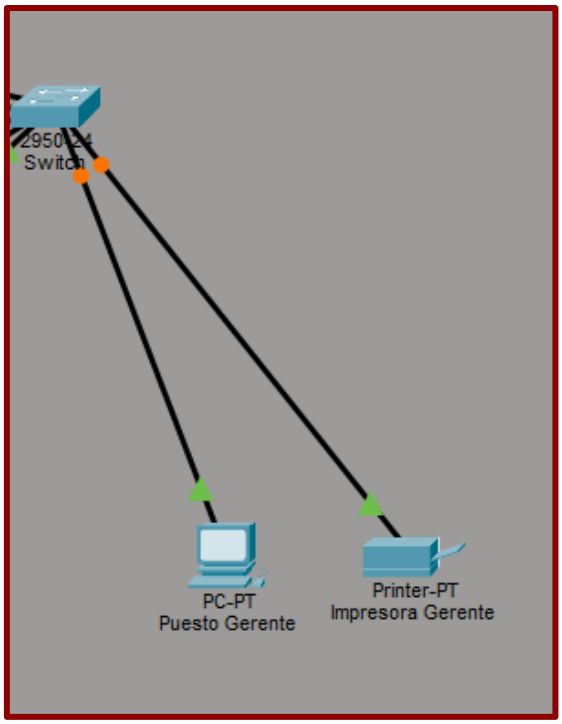
Equipo del Gerente:



Impresora del Gerente:

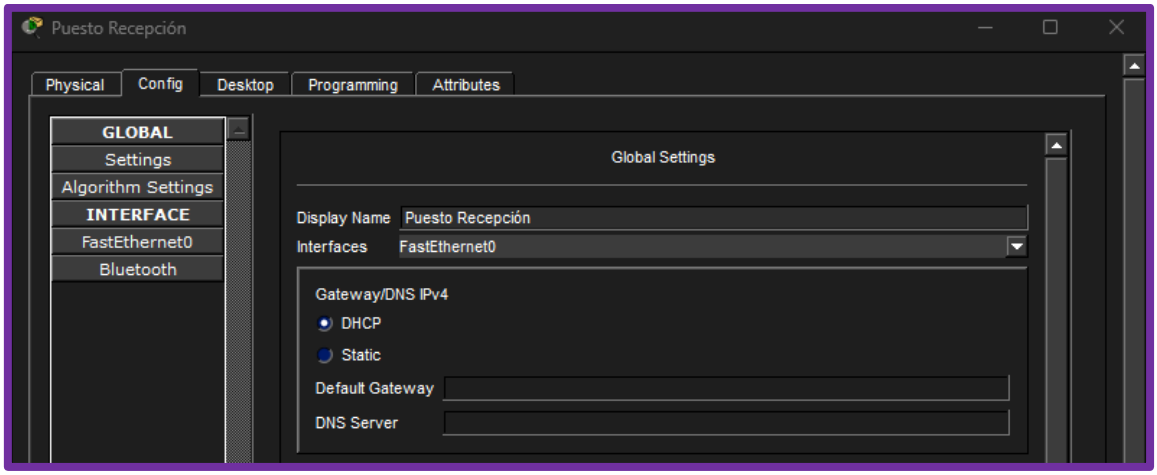






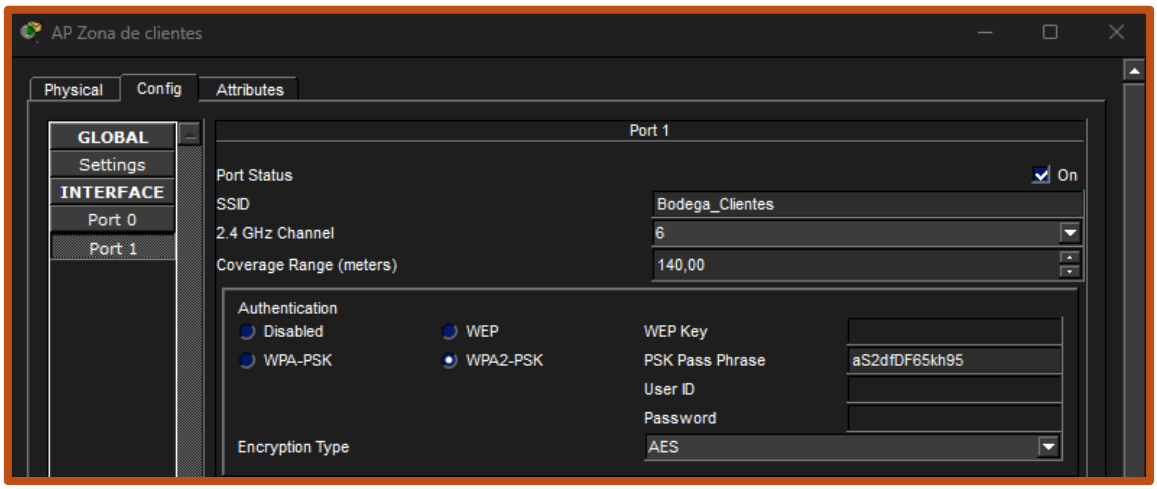
VLAN 30

Equipo del Puesto de Recepción:



VLAN 40

Punto de Acceso para Clientes:



Configurar las VLAN

En la configuración CLI (Consola de comandos) escribimos:

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Administracion
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Gerencia
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name Recepcion
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name ClientesWiFi
Switch(config-vlan)#exit
Switch(config)#
```

Con esto creamos las VLAN 10 (Administración), 20 (Gerencia), 30 (Recepción) y 40 (ClientesWiFi).

Primero configuramos el puerto del Servidor (Fa/02):

```
Switch(config)#interface FastEthernet0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
```

De los puertos Fa0/3 al 0/6 (Los 4 equipos) y el puerto Fa0/7 (La impresora) forman parte de la VLAN 10.

```
Switch>enable
Switch#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range FastEthernet0/3 - 6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
```

```
Switch(config)#interface FastEthernet0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
```

El puerto Fa/08 (Equipo del gerente) y el Fa/09 (Impresora del gerente) forman parte de la VLAN 20.

```
Switch(config)#interface FastEthernet0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface FastEthernet0/9
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
```

El puerto Fa0/10 (Equipo del recepcionista) forma parte de la VLAN 30.

```
Switch(config)#interface FastEthernet0/10
Switch(config-if)#Switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#
```

El puerto Fa0/11 (Punto de acceso para clientes) forma parte de la VLAN 40.

```
Switch(config)#interface FastEthernet0/11
Switch(config-if)#Switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#exit
Switch(config)#
```

Configuración del Router

Primero abrimos la conexión entre Router y Switch:

```
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#
```

Configuramos el Gateway del VLAN 10:

```
Router(config)#interface GigabitEthernet0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#
```

Configuramos el Gateway del VLAN 20:

```
Router(config)#interface GigabitEthernet0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#
```

Configuramos el Gateway del VLAN 30:

```
Router(config)#interface GigabitEthernet0/0.30
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#
```

Configuramos el Gateway del VLAN 40:

```
Router(config)#interface GigabitEthernet0/0.40
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up
encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

Creamos las Pools DHCP (reservando las 20 primeras IP para los Hosts e Impresoras).

VLAN 10:

```
ip dhcp excluded-address 192.168.10.1 192.168.10.20

ip dhcp pool VLAN10
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
exit
```

VLAN 20:

```
ip dhcp excluded-address 192.168.20.1 192.168.20.20

ip dhcp pool VLAN20
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
exit
```

VLAN 30:

```
ip dhcp excluded-address 192.168.30.1 192.168.10.20

ip dhcp pool VLAN30
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 8.8.8.8
exit
```

VLAN 40:

```
ip dhcp excluded-address 192.168.40.1 192.168.40.20

ip dhcp pool VLAN10
network 192.168.40.0 255.255.255.0
default-router 192.168.40.1
dns-server 8.8.8.8
exit
```

```
Router#show ip dhcp pool

Pool VLAN10 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 4
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.10.1         192.168.10.1     - 192.168.10.254    0    / 4     / 254

Pool VLAN20 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 4
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.20.1         192.168.20.1     - 192.168.20.254    0    / 4     / 254

Pool VLAN30 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 4
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.30.1         192.168.30.1     - 192.168.30.254    0    / 4     / 254

Pool VLAN40 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 4
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.40.1         192.168.40.1     - 192.168.40.254    0    / 4     / 254
```

De esta forma se asignan las IP de los equipos con DHCP automáticamente.

Pruebas de funcionamiento

Prueba 1:

Desde la consola del equipo del Puesto 1, intentamos comunicarnos con el equipo del Recepcionista:

IP del equipo del Recepcionista:

Consola del equipo del Puesto 1:

```
C:\>ping 192.168.30.21

Pinging 192.168.30.21 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Resultado:

Como el equipo del Puesto 1 (de la VLAN 10) y el equipo del Recepcionista (VLAN 30) son de VLAN distintas, al hacer ping entre ellos, no devuelven la conexión.

Prueba 2:

Desde la consola del equipo del Puesto 2, intentamos comunicarnos con el equipo del Puesto 3:

IP del equipo del Puesto 3:

Consola del equipo del Puesto 2:

```
Pinging 192.168.10.23 with 32 bytes of data:

Reply from 192.168.10.23: bytes=32 time<1ms TTL=128
Reply from 192.168.10.23: bytes=32 time<1ms TTL=128
Reply from 192.168.10.23: bytes=32 time<1ms TTL=128
Reply from 192.168.10.23: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.23:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

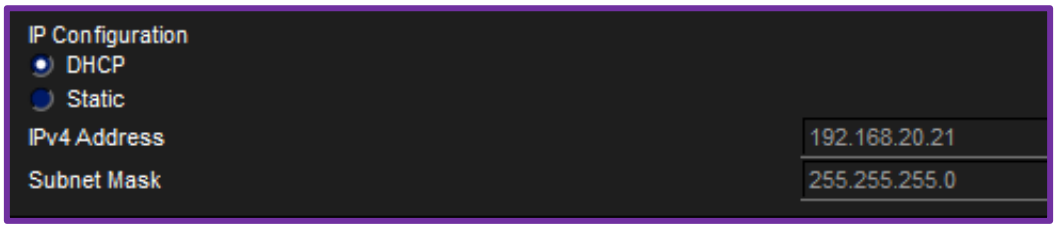
Resultado:

Como el equipo del Puesto 2 y del Puesto 3 son de la misma VLAN al hacer ping entre ellos, si consiguen comunicarse.

Prueba 3:

Desde la consola del equipo del Puesto 4, intentamos comunicarnos con el equipo del Gerente:

IP del equipo del Gerente:



Consola del equipo del Puesto 4:

```
C:\>ping 192.168.20.21

Pinging 192.168.20.21 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Resultado:

Como el equipo del Puesto 4 (de la VLAN 10) y el equipo del Gerente (VLAN 20) son de VLAN distintas, al hacer ping entre ellos, no devuelven la conexión.